

[Comparison of the efficacy and safety of praziquantel administered in single dose of 40 versus 60 mg/kg for treating urinary schistosomiasis in Mauritania].

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During the last twenty years, praziquantel (PZQ) was the drug of choice for the treatment of schistosomiasis in the majority of national programs. However, a lower rate of cure had been significantly noted on the left bank of the Senegal River. To explain this unusual rate of cure, the assumption of a possible resistance to the drug as well as under-dosing was considered. With an aim of testing this hypothesis of underdosing, we compared the amount of a single dose of 60 mg/kg of PZQ versus the standardized dose of 40 mg/kg used in curing urinary schistosomiasis in Mauritania. One hundred and fifty-one children aged from 10 to 19 years, including 77 in the group of 60 mg/kg and 74 in the group of 40 mg/kg, were included in the study. The rates of cure were respectively 64.8% for 60 mg/kg and 67.5% for 40 mg/kg three weeks after the administration of the treatment without statistically significant difference. For the majority of the patients, the drug was well tolerated and no serious adverse events were noted; however, clinical signs in the form of abdominal pain associated or not with diarrhea and vomiting were noted. Praziquantel remains an effective and well-tolerated drug: the amount of 40 mg/kg of body weight can still be maintained for the treatment of schistosomiasis in Mauritania.

Urogenital schistosomiasis in three different water access in the Senegal river basin: prevalence and monitoring praziquantel efficacy and re-infection levels.

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Urogenital schistosomiasis is a neglected tropical disease most prevalent in sub-Saharan Africa. In the Senegal river basin, the construction of the Diama dam led to an increase and endemicity of schistosomiasis. Since 2009, praziquantel has frequently been used as preventive chemotherapy in the form of mass administration to Senegalese school-aged children without monitoring of the treatment efficacy and the prevalence after re-infection. This study aims to determine the current prevalence of urogenital schistosomiasis (caused by *Schistosoma haematobium*), the efficacy of praziquantel, and the re-infection rates in children from five villages with different water access. The baseline prevalence of *S. haematobium* was determined in August 2020 in 777 children between 5 and 11 years old and a single dose of praziquantel (40 mg/kg) was administered to those positive. The efficacy of praziquantel and the re-infection rates were monitored 4 weeks and 7 months after treatment, respectively, in 226 children with a high intensity of infection at baseline. At the baseline, prevalence was low among children from the village of Mbane who live close to the Lac de Guiers (38%), moderate among those from the villages of Dioundou and Khodit, which neighbor the Doue river (46%), and very high at Khodit (90.6%) and Guia (91.2%) which mainly use an irrigation canal. After treatment, the observed cure rates confirmed the efficacy of praziquantel. The lowest cure rate (88.5%) was obtained in the village using the irrigation canal, while high cure rates were obtained in those using the lake (96.5%) and the river (98%). However, high egg reduction rates (between 96.7 and 99.7%) were obtained in all the villages. The re-infection was significantly higher in the village using the canal (42.5%) than in the villages accessing the Lac de Guiers (18.3%) and the Doue river (14.8%). Praziquantel has an impact on reducing the prevalence and intensity of urogenital schistosomiasis. However, in the Senegal river basin, *S. haematobium* remains a real health problem for children living in the villages near the irrigation canals, despite regular treatment, while prevalence is declining from those frequenting the river and the Lac de Guiers. Trial registration ClinicalTrials.gov, NCT04635553. Registered 19 November 2020 retrospectively registered, <https://www.clinicaltrials.gov/ct2/show/study?term=NCT04635553>

gov/ct2/show/NCT04635553?cntry=SN&draw=2&rank=4.

Prevalence and factors associated with persistent transmission of *Schistosoma haematobium* among primary school children after five rounds of mass drug administration using praziquantel: A cross sectional study in Mkuranga district, Tanzania.

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Despite a human schistosomiasis control programme through praziquantel mass drug administration (MDA) between 2011 and 2015, there was still persistent transmission among primary school children (PSC) in Mkuranga district, Tanzania. Our cross-sectional study was conducted among 396 PSC who provided urine for diagnosis of *Schistosoma haematobium* infection. Observations were conducted to determine PSC water contact activities. Logistic regression was used to test association between dependent and independent variables. We found MDA uptake among PSC as 72.5%, and the prevalence of *Schistosoma haematobium* infection 5.8%. The risk of infection increased among PSC engaged in fetching water and adjusted odds ratio (AOR) for swimming, bathing, fishing, crossing ponds and paddy fields were 0.123, 0.166, 0.232, 0.202 and 0.093 respectively. Thus we conclude that multiple water contact activities and low participation in MDA is responsible for persistent *Schistosoma* transmission.

Persistence of *Schistosoma haematobium* transmission among school children and its implication for the control of urogenital schistosomiasis in Lindi, Tanzania.

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Despite twelve rounds of school-based preventive chemotherapy for schistosomiasis in endemic areas of Tanzania such as Mtama district, Lindi: the burden of *Schistosoma haematobium* infection has remained highly conceivable due to re-infections. The factors associated with continuity of *S. haematobium* transmission in Mtama district, Lindi have not been fully established. This study investigated the burden and factors contributing to the ongoing transmission of *S. haematobium* infection in the endemic district of Mtama, Lindi. A quantitative cross-sectional survey was carried out among 649 school-age children in the Mtama district to determine the burden and factors associated with continuity of *S. haematobium* infection transmission. A single urine specimen was obtained from each pupil and tested for macro- and microhaematuria, presence of *S. haematobium* ova, as well as intensity of infection; this was complemented with a survey of *Bulinus* spp snail intermediate hosts and their infectivity. A structured questionnaire was employed to gather information on individual and environmental risk factors for *S. haematobium* transmission. Summary statistics were computed for individual variables; while a univariate and multivariate logistic regression analysis was performed to assess the association between risk factors with *S. haematobium* infection. Prevalence of *S. haematobium* infection by macro- and microhaematuria was 13.1% and 46.2% respectively. The prevalence of *S. haematobium* ova was 52.7%; intensity of infection was light in 53.1%, and heavy in 46.9%. Snail intermediate hosts were *Bulinus globosus* and *B. nasutus*, whose infectivity was 2.2% and 1.3%, respectively. Among the assessed risk factors, long residency (10-13 years) in the area was a significant risk factor for the continuity of *S. haematobium* transmission (AOR: 21.79, 95% CI: 1.37-346.4). The observed 52.7% prevalence of *S. haematobium* infection represents unacceptably high prevalence after 12 rounds of preventive chemotherapy.

Therefore, an urgent need for the implementation of integrated multiple control interventions in the Mtama district; is considered to be imperative.

Urogenital schistosomiasis infection prevalence targets to determine elimination as a public health problem based on microhematuria prevalence in school-age children.

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Recent research suggests that schistosomiasis targets for morbidity control and elimination as a public health problem could benefit from a reanalysis. These analyses would define evidence-based targets that control programs could use to confidently assert that they had controlled or eliminated schistosomiasis as a public health problem. We estimated how low *Schistosoma haematobium* infection levels diagnosed by urine filtration in school-age children should be decreased so that microhematuria prevalence was at, or below, a "background" level of morbidity. Data obtained from school-age children in Burkina Faso, Mali, Niger, Tanzania, and Zambia who participated in schistosomiasis monitoring and evaluation cohorts were reanalyzed before and after initiation of preventive chemotherapy. Bayesian models estimated the infection level prevalence probabilities associated with microhematuria thresholds $\leq 10\%$, 13% , or 15% . An infection prevalence of 5% could be a sensible target for urogenital schistosomiasis morbidity control in children as microhematuria prevalence was highly likely to be below 10% in all surveys. Targets of 8% and 11% infection prevalence were highly likely to result in microhematuria levels less than 13% and 15% , respectively. By contrast, measuring heavy-intensity infections only achieves these thresholds at impractically low prevalence levels. A target of 5% , 8% , or 11% urogenital schistosomiasis infection prevalence in school-age children could be used to determine whether a geographic area has controlled or eliminated schistosomiasis as a public health problem depending on the local background threshold of microhematuria.

Mapping *Schistosoma haematobium* for Novel Interventions against Female Genital Schistosomiasis and Associated HIV Risk in KwaZulu-Natal, South Africa.

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Women with female genital schistosomiasis (FGS) have been found to have genital symptoms and a three-fold higher risk of HIV infection. Despite WHO recommendations, regular antischistosomal mass drug administration (MDA) has not yet been implemented in South Africa possibly because of the lack of updated epidemiological data. To provide data for future prevention efforts against FGS and HIV, this study explored *Schistosoma haematobium* prevalence in girls and young women and the effects of antischistosomal MDA, respectively. Urinary schistosomiasis and genital symptoms were investigated in 70 randomly selected secondary schools in three districts within KwaZulu-Natal and 18 primary schools. All study participants were treated for schistosomiasis, and schools with the highest urinary prevalence were followed up after 1 and 4 years of MDA. At baseline, urine analysis data showed that most schools were within the moderate-risk prevalence category where biennial antischistosomal MDA is recommended, as per WHO guidelines. Young women had high prevalence of genital symptoms (36%) after correcting for sexually transmitted infections. These symptoms may be caused by infection with schistosomes. However, FGS cannot be diagnosed by urine analysis alone. In KwaZulu-Natal rural schools, this study suggests that antischistosomal MDA with praziquantel could prevent genital

symptoms in more than 200,000 young women. Furthermore, it is feasible that more than 5,000 HIV infections could be prevented in adolescent girls and young women by treatment and prevention of FGS.

Drugs for treating urinary schistosomiasis.

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Urinary schistosomiasis is caused by an intravascular infection with parasitic *Schistosoma haematobium* worms. The adult worms typically migrate to the venous plexus of the human bladder and excrete eggs which the infected person passes in their urine. Chronic infection can cause substantial morbidity and long-term complications as the eggs become trapped in human tissues causing inflammation and fibrosis. We summarised evidence of drugs active against the infection. This is new edition of a review first published in 1997. To evaluate the efficacy and safety of drugs for treating urinary schistosomiasis. We searched the Cochrane Infectious Diseases Group Specialized Register, MEDLINE, CENTRAL, EMBASE and LILACS and reference lists of articles up to 23 May 2014. Randomized controlled trials (RCTs) of antischistosomal drugs and drug combinations compared to placebo, no intervention, or each other. Two researchers independently screened the records, extracted the data and assessed risk of bias. The primary efficacy outcomes were parasitological failure (defined as the continued presence of *S. haematobium* eggs in the urine at time points greater than one month after treatment), and percent reduction of egg counts from baseline. We presented dichotomous data as risk ratios (RR), and continuous data as mean difference (MD), alongside their 95% confidence intervals (CIs). Where appropriate we combined trials in meta analyses or tables. We assessed the quality of evidence using the GRADE approach. We included 30 RCTs enrolling 8165 participants in this review. Twenty-four trials were conducted in children in sub-Saharan Africa, and 21 trials were over 20 years old. Many studies were assessed as being at unclear risk of bias due to inadequate descriptions of study methods. Praziquantel On average, a single 40 mg/kg dose of praziquantel reduced the proportion of people still excreting eggs in their urine by around 60% compared to placebo at one to two months after treatment (treatment failure: RR 0.42, 95% CI 0.29 to 0.59, 864 participants, seven trials, high quality evidence). The proportion of people cured with praziquantel varied substantially between trials, from 22.5% to 83.3%, but was higher than 60% in five of the seven trials. At one to two months following praziquantel treatment at 40 mg/kg, the mean number of schistosome eggs in the urine was reduced by over 95% in five out of six trials (678 participants, six trials, high quality evidence). Splitting praziquantel 40 mg/kg into two doses over 12 hours probably has no benefits over a single dose, and in a single trial of 220 participants the split dose caused more vomiting (RR 0.5, 95% CI 0.29 to 0.86) and dizziness (RR 0.39, 95% CI 0.16 to 0.94). Metrifonate A single dose of metrifonate 10 mg/kg reduced egg excretion (210 participants, one trial, at eight months), but was only marginally better than placebo at achieving cure at one month (RR 0.83, 95% CI 0.74 to 0.94, 142 participants, one trial). In a single trial comparing one, two and three doses, the absolute number of participants cured improved from 47% after one dose to 81% after three doses (93 participants, one trial, low quality evidence). Two small trials compared 40 mg/kg single dose praziquantel with two or three doses of 10 mg/kg metrifonate and found no clear evidence of differences in cure (metrifonate 2 x 10 mg/kg at one month: RR 1.03, 95% CI 0.8 to 1.34, 72 participants, one trial; metrifonate 3 x 10 mg/kg at three months: RR 0.33, 95% CI 0.07 to 1.57, 100 participants, one trial. In one trial both drugs performed badly and in one trial both performed well. Other drugs Three trials have evaluated the antimalarial artesunate; with inconsistent results. Substantial antischistosomal effects were only seen in one of the three trials, which was at unclear risk of bias due to poor reporting of the trial methods. Similarly, another anti-malarial mefloquine has been evaluated in two small trials with inconsistent effects. Adverse events were described as mild for all evaluated drugs, but adverse event

monitoring and reporting was generally of low quality. Praziquantel 40 mg/kg is the most studied drug for treating urinary schistosomiasis, and has the strongest evidence base. Potential strategies to improve future treatments for schistosomiasis include the combination of praziquantel with metrifonate, or with antimalarial drugs with antischistosomal properties such as artesunate and mefloquine. Evaluation of these combinations requires rigorous, adequately powered trials using standardized outcome measures.

Urinary schistosomiasis among schoolchildren in Yemen: prevalence, risk factors, and the effect of a chemotherapeutic intervention.

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Schistosomiasis is one of the most important public health problems in Yemen. The prevalence of urinary schistosomiasis varies considerably across different parts of Yemen and was estimated to be 10% among schoolchildren in Sana'a. Praziquantel (PZQ) is highly effective against all five major human species of schistosomes. The aim of the present work was to estimate the prevalence of urinary schistosomiasis, describe the risk factors associated with its endemicity, and implement and assess a chemotherapeutic intervention using PZQ in a village in Yemen. The sample included 696 schoolchildren from a village in Abyan Governorate. During the baseline school survey, personal, sociodemographic, and environmental data, and data on practices in relation to water contact were collected from each study participant using a predesigned structured questionnaire. Urine samples from each participant were examined for macrohematuria and the presence of *Schistosoma haematobium* eggs. The chemotherapeutic intervention was assessed 3 and 6 months after the treatment and certain indicators were calculated. The prevalence of *S. haematobium* was 18.1%. The main significant risk factors were male sex; proximity of houses to water ponds; and using pond water for swimming, agricultural activities, and for bathing in houses. PZQ treatment reduced the prevalence of infection and decreased the prevalence of high-intensity infection. Survival analysis showed that the probability of residual infection also dropped after the treatment intervention. Male sex and using pond water for various activities were the main significant risk factors associated with urinary schistosomiasis. PZQ is still a cornerstone drug in reducing or eliminating morbidity associated with schistosomiasis infection. Health education programs tailored for the community are required for the control and prevention of urinary schistosomiasis. To address schoolchildren, school curricula should include lessons about urinary schistosomiasis.

A novel sustained-release agent based on disulfide-induced recombinant collagen hydrogels for the prevention and treatment of *Schistosoma* infections.

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Schistosomiasis is commonly managed using the praziquantel, but it is only effective against adult worms and duration of action is short. Liver fibrosis will worsen if eggs are still present after stopping treatment. Therefore, this study aimed to develop a sustained drug release system for effectively preventing and treating schistosomiasis. A disulfide bond-induced three-dimensional (3D) recombinant collagen hydrogel was developed for sustained praziquantel release. Three collagen sequences were developed based on the sequence for Scl2 of *S. pyogene*, with different substitutions of residues for cysteine (S-VCL-S1, S-VCL-S2, and S-VCL-S3). Their properties were tested. Mice were infected with *Schistosoma japonicum* cercariae and treated either with praziquantel collagen hydrogel or niclosamide collagen hydrogel. The

worm-killing effect was examined. The application of hydrogel-niclosamide on the skin for 24 h effectively prevented Schistosoma cercariae from infecting mice and showed 70.95% and 81.73% reduction in the number of eggs and worms, respectively. The combined use of niclosamide and anti-cercariae cream showed 100% protection after 24 h. The hydrogel-praziquantel also showed a 100% reduction of worms and eggs after 24 h of subcutaneous injection. The subcutaneous injection of praziquantel after 28 days of infection showed 95.19% and 80.12% reduction of worm and egg counts, respectively, and the development of larvae was significantly slowed down. Liver analysis showed no infection after 7 days of treatment. These results suggest developing a novel type of sustained-release agent against schistosomiasis based on the recombinant collagen hydrogel that provides a potential new treatment for schistosomiasis. **IMPORTANCE** This study introduces a new way for treating schistosomiasis: a special collagen hydrogel that gradually releases medication to treat schistosomiasis effectively. This innovation provides a promising way to treat schistosomiasis. It represents a significant step forward in the fight against this disease and offers hope for more effective and safer treatments in the future.

Schistosomiasis Chemotherapy, Chemoprevention, and Vaccines: History, Progress, and Priorities.

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Schistosomiasis is a major human disease of public health importance. Freshwater snails serving as intermediary hosts and human interaction with surface water tainted by feces or urine are both necessary components of the transmission cycle. *Schistosoma haematobium*, *Schistosoma mansoni*, and *Schistosoma japonicum* are the primary pathogen species. Over 250 million individuals are infected globally, according to the World Health Organization, causing significant morbidity and an estimated loss of 1.9 million disability-adjusted life years, a number that is probably underestimated. Immunological protection is slowly built up through complex immunological systems, although innate factors also play a role. Chronic schistosomiasis affects mainly individuals residing in poor rural area. Vaccination is considered as one of the most sustainable options for the control of any pathogen, but schistosomiasis vaccine for humans or animals is not available till now despite the discovery of numerous potentially promising schistosome vaccine antigens. To provide an overview of the schistosomiasis chemotherapy, chemoprevention, and vaccines history and progress. Review article. PubMed, ISI Web of Science, Science Direct, and the World Health Organization database. Favorably praziquantel (PZQ) is a medication with excellent chemopreventive treatment compliance. Due to the extensive usage of PZQ, there is a great deal of debate surrounding the emergence of drug resistance. PZQ is effective against all species of schistosomes, schistosomiasis prevalence has remained largely unaffected, due to reinfection in high transmission areas and growing juvenile worms that were not affected by the drug, even though the need for a schistosomiasis vaccine is even more pressing.